

Qatar-1b

R-2643

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250917 · created 2026-08-29T01:35:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460935.8094 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.085 +/- 0.055
Transit depth (Rp/Rs)^2	0.73 +/- 0.93 [%]
Orbital Inclination (inc)	88.95 +/- 2.85
Airmass coefficient 1 (a1)	2.98 +/- 1.76
Airmass coefficient 2 (a2)	-0.71 +/- 0.62
Scatter in the residuals of the lightcurve fit is	58.26 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.0755 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.085 $\pm$ 0.055 vs 0.1463 (deviation 1.11 σ)
Duration fit vs published	0.0755 d vs 0.0692 d (deviation 0.88 σ)
Mid-time O-C	7.7 min
Depth SNR	1.5
Residual scatter	58.26 %

Light curve

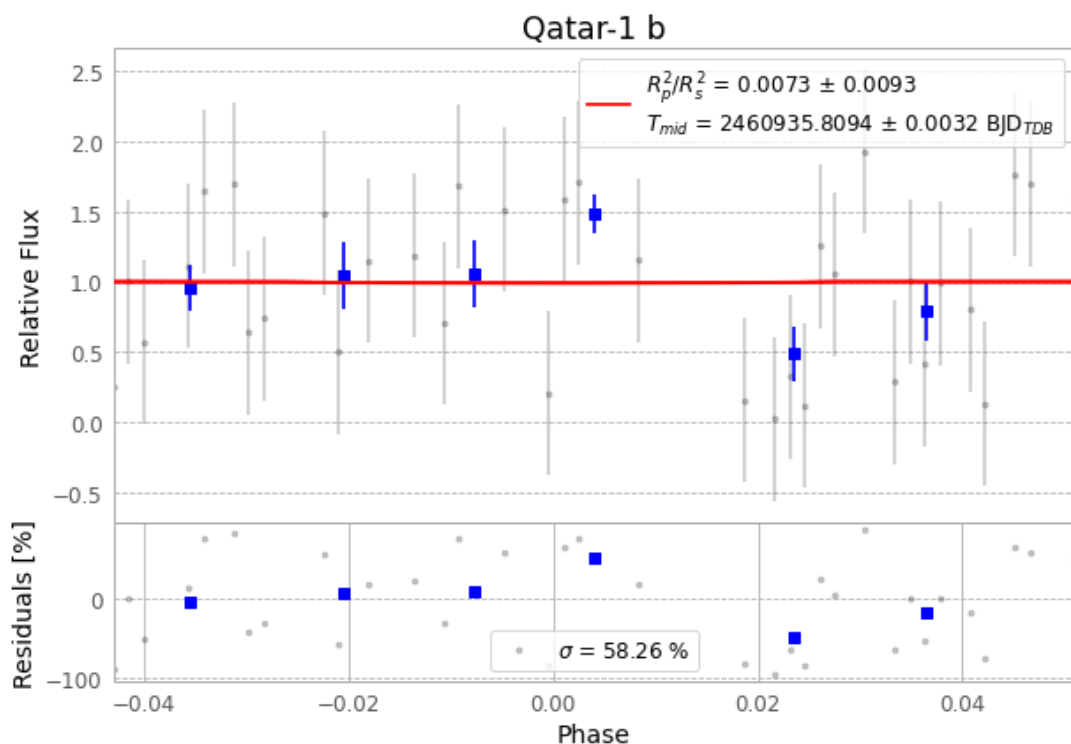


Figure 1 — FinalLightCurve_Qatar-1 b_2025-09-16.png (SHA-256 ae3d00bdf5a0dd45...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [570, 144], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a9af248e8852e45e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-1250917055116.FITS → Qatar-1250917090615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250917.log (a68a3f24de3f...), FinalLightCurve_Qatar-1 b_2025-09-16.png (ae3d00bdf5a0...), FinalLightCurve_Qatar-1 b_2025-09-16.csv (d417575acf04...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 01:35Z.